Project Plan

Ontology-Grounded LLMs for Detecting Breast Cancer Misinformation in Patient Conversations

NLP Course Assignment

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1. Project Overview

Objective: Build a system that detects misinformation in breast cancer patient conversations by decomposing patient statements into structured claims and verifying them against a purpose-built Breast Cancer Misinformation Ontology (BCMO).

Central Hypothesis: Explicit ontology grounding makes LLM-based misinformation detection more accurate, robust, and explainable than relying on the LLM's parametric knowledge or retrieval alone.

Deliverables: 1. Breast Cancer Misinformation Ontology (BCMO) in JSON/OWL format 2. Claim decomposition pipeline (patient text → structured triples) 3. Ontology-grounded verification system 4. Comparative evaluation: Ontology-grounded vs. LLM-only vs. RAG 5. Final paper with results and analysis

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2. Research Questions

RQ	Question	How We Answer It
RQ1	Does domain fine-tuning improve concept/relation extraction?	Compare base vs. fine-tuned LLM on claim extraction task
RQ2	Does ontology grounding beat LLM-only?	Compare system with/without BCMO on misinfo detection
RQ3	Does ontology provide value beyond RAG?	Compare ontology-grounded vs. standard RAG retrieval
RQ4	Does grounding improve robustness?	Test on paraphrased/implicit misinformation examples
RQ5	Does grounding produce faithful explanations?	Human evaluation of explanation quality

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3. Data Sources & Access Plan

3.1 MedQuAD — Cancer.gov Subset (PRIMARY DATA) Available Now

- **What:** 116 XML files with cancer QA pairs, UMLS CUIs, semantic types, question types
- **Location:** `MedQuAD/1_CancerGov_QA/` (already cloned)
- **License:** CC BY 4.0
- **Use:** Ontology concept extraction, domain adaptation fine-tuning, claim templates

3.2 Reddit r/breastcancer Posts Accessible via API

- **What:** Patient-generated discussions about breast cancer (informal language)
- **Access:** Reddit API (free developer account)
- **Use:** Domain language adaptation, test examples of patient misinformation
- **Plan:** Collect 5,000-10,000 posts from r/breastcancer, r/cancer, r/HER2positive

3.3 Cancer-Myth Benchmark Request from Authors

- **What:** 585 cancer questions with false presuppositions + 150 controls
- **Paper:** Zhu et al., arXiv:2504.11373 (USC)
- **Use:** Adversarial evaluation benchmark
- **Fallback:** If unavailable, construct 50-100 breast-cancer-specific questions with false presuppositions ourselves (using MedQuAD knowledge + clinical guidelines)

3.4 NCI Thesaurus / UMLS Free with License

- **What:** Structured medical terminology, breast cancer concepts, treatment hierarchies
- **Access:** NLM UMLS license (free, 1-2 day approval)
- **Use:** Foundation for BCMO ontology construction

3.5 Self-Constructed Evaluation Set (OUR CONTRIBUTION)

- **What:** 100-200 breast cancer patient statements annotated for misinformation
- **Source:** Derived from Reddit posts + paraphrased MedQuAD content + synthetic examples
- **Annotation:** Binary (misinfo/not) + claim-level labels + explanation

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4. System Architecture

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Patient Statement


  Claim Decomposer    (LLM extracts subject-relation-object triples)
   (Fine-tuned)




  Entity Linker       (Maps entities to BCMO concepts)
   (ScispaCy/UMLS)




  Ontology Verifier   (Checks claims against BCMO relations)
   (BCMO + Rules)




  Evidence Scorer     (Scores: supported / contradicted / unsupported)
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```
+ Explainer


VERDICT + EXPLANATION
(e.g., "MISLEADING: HER2-targeted therapy and chemotherapy
 are often used together, not as alternatives")
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5. Implementation Plan (Phase by Phase)

Phase 1: Data Preparation & Ontology Construction

Goal: Build BCMO and prepare all datasets

Tasks: - [ ] Parse MedQuAD Cancer.gov XML files → extract breast-cancer concepts, relations, treatments - [ ] Collect Reddit r/breastcancer posts via API (5K-10K posts) - [ ] Apply for UMLS license, download NCI Thesaurus breast cancer subset - [ ] Construct BCMO v1.0: - Define concept classes (16 classes from proposal) - Define relation types (12 relations from proposal) - Populate with breast cancer concepts from MedQuAD + NCI - Add negation/uncertainty/strength modifiers - [ ] Create evaluation dataset: 100-200 annotated patient statements

Output: `bcmo.json` (ontology), `eval_dataset.csv` (annotated test set), `reddit_posts.jsonl` (patient language)

Phase 2: Claim Decomposition Pipeline

Goal: Convert patient text into structured claims

Tasks: - [ ] Design claim format: `(subject, relation, object, strength, negation)` - Example: `("HER2-positive", "incompatible_with", "chemotherapy", "absolute", false)` → MISLEADING - [ ] Implement prompted claim extraction using base LLM (Llama-3-8B or GPT-4 API) - [ ] Fine-tune a smaller model on MedQuAD-derived claim extraction examples (LoRA) - [ ] Evaluate: precision/recall of extracted claims vs. manual annotation - [ ] Compare base vs. fine-tuned model (answers RQ1)

Output: `claim_extractor.py`, fine-tuned model weights, extraction evaluation results

Phase 3: Ontology-Grounded Verification

Goal: Build the core verification system

Tasks: - [ ] Implement entity linking (ScispaCy or QuickUMLS → BCMO nodes) - [ ] Implement claim verification logic: - Claim matches BCMO relation → SUPPORTED - Claim contradicts BCMO relation → CONTRADICTED (misinformation) - Claim has no BCMO match → UNSUPPORTED (flag for review) - Claim overstates strength ("always" vs "can") → MISLEADING - [ ] Build RAG baseline (retrieve relevant PubMed/Cancer.gov passages, ask LLM to verify) - [ ] Build LLM-only baseline (ask LLM directly without ontology or retrieval) - [ ] Compare all three systems on evaluation set (answers RQ2, RQ3)

Output: `ontology_verifier.py`, `rag_baseline.py`, `llm_baseline.py`, comparison results

Phase 4: Robustness & Explanation Evaluation

Goal: Test robustness and explanation quality

Tasks: - [ ] Create paraphrased test set (same misinformation, different wording) - [ ] Create implicit misinformation examples (false presuppositions in questions) - [ ] Create counterfactual claims (swap entities to test if model relies on shortcuts) - [ ] Evaluate all three systems on robustness set (answers RQ4) - [ ] Human evaluation of explanation faithfulness: - Does the explanation cite the correct ontology relation? - Is the explanation consistent with the verdict? - Rate 1-5 on clarity and helpfulness (answers RQ5)

Output: Robustness results, explanation evaluation scores, qualitative examples

Phase 5: Paper Writing & Presentation

Goal: Document findings

Tasks: - [ ] Write Introduction + Related Work (using our research docs) - [ ] Write Methodology (system architecture, BCMO design, baselines) - [ ] Write Results (tables comparing systems across RQ1-RQ5) - [ ] Write Discussion (what worked, limitations, future work) - [ ] Prepare presentation slides

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6. Technical Stack

Component	Tool	Why
Language	Python 3.10+	Standard for NLP
LLM (base)	Llama-3-8B-Instruct	Open-weight, fine-tunable, strong baseline
LLM (API)	GPT-4o or Claude	For claim extraction comparison / annotation assist
Fine-tuning	HuggingFace PEFT + LoRA	Efficient, runs on single GPU
Entity linking	ScispaCy + UMLS	Medical NER + concept normalization
Ontology storage	JSON (lightweight) or NetworkX (graph queries)	Simple, no DB setup needed
RAG	LangChain + FAISS + BioSentVec	Standard RAG pipeline
Evaluation	scikit-learn (F1, precision, recall)	Standard metrics
Data processing	pandas, xml.etree, praw (Reddit)	Data parsing + collection

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7. Expected Results & Contributions

Quantitative (Expected)

System	Misinfo Detection F1	Robustness (paraphrase)	False Positive Rate
LLM-only	~0.55-0.65	Low (drops on paraphrase)	High
RAG	~0.65-0.75	Medium	Medium
Ontology-grounded	**~0.75-0.85**	**High (stable)**	**Low**

Qualitative

- Ontology-grounded system provides traceable explanations citing specific BCMO relations
- LLM-only system gives vague "this may not be accurate" responses
- RAG system retrieves relevant passages but may not catch implicit contradictions

Novel Contributions

1. **BCMO** — first breast-cancer-specific misinformation ontology with negation/uncertainty
2. **Structured claim verification** for patient-generated text (vs. existing work on scientific literature)
3. **Comparative analysis** of ontology vs. RAG vs. LLM-only for medical misinformation
4. **Evaluation dataset** of annotated breast cancer patient misinformation

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8. Risk Mitigation

Risk	Mitigation
Cancer-Myth dataset not available	Build own 50-100 adversarial questions from MedQuAD knowledge
Reddit API rate limits	Collect data early; cache locally; 5K posts is sufficient
LLM fine-tuning compute limits	Use LoRA (0.5% params); Colab Pro or university GPU
Ontology too small to cover all cases	Focus on top-20 breast cancer misinformation topics (HER2, chemo, hormone therapy, BRCA, staging)
Human evaluation resources limited	Annotate 50-100 examples ourselves; use LLM-as-judge as supplement

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9. File Structure

```
nlp/
  data/
      MedQuAD/                    # Cloned (done)
      reddit_posts/               # Collected via API
      eval_dataset.csv            # Our annotated test set
      bcmo.json                   # Ontology
  src/
      data_prep/
          parse_medquad.py        # Extract breast cancer content
          collect_reddit.py       # Reddit API collector
      ontology/
          build_bcmo.py           # Ontology construction
          bcmo_queries.py         # Query/verify functions
      pipeline/
          claim_extractor.py      # Decompose text → claims
          entity_linker.py        # Link to BCMO
          verifier.py             # Ontology-grounded verification
      baselines/
          llm_only.py             # LLM-only baseline
          rag_baseline.py         # RAG baseline
      evaluation/
           evaluate.py            # Metrics computation
           robustness_tests.py    # Paraphrase/implicit tests
  notebooks/
      01_data_exploration.ipynb
      02_ontology_construction.ipynb
      03_claim_extraction.ipynb
      04_verification_experiments.ipynb
      05_results_analysis.ipynb
  paper/
```

```
    main.tex (or .docx)
docs/
    topic4_summary.md
    topic4_related_work.md
    topic4_data_analysis.md
    project_plan.md             # This file
```

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10. What We Submit

1. **Paper** (8-12 pages): Introduction, Related Work, Methodology, Experiments, Results, Discussion
2. **Code repository**: All source code, notebooks, and data processing scripts
3. **BCMO ontology**: The constructed ontology (shareable artifact)
4. **Evaluation dataset**: Annotated breast cancer misinformation examples
5. **Presentation**: 10-15 minute summary of approach and findings